SENTINEL AI

AI-Powered Digital Public Safety Intelligence Platform

Defeating Counterfeiting, Fraud & Digital Arrest Scams

Theme: Smart Cities · Public Safety · Digital Trust · Geospatial Law Enforcement

Hackathon Proposal Document

Problem Statement · Proposed Solution · Feature Architecture · System Design

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1. Problem Statement

India registered 1.14 million cybercrime complaints in 2023, a rise of roughly 60% over the previous year, and the trend has continued to steepen. Among the most damaging patterns is the 'digital arrest' scam, in which fraudsters impersonating agencies such as the CBI, Enforcement Directorate, or Customs trap victims in prolonged psychological hostage situations conducted over video call. According to the Ministry of Home Affairs, such scams defrauded citizens of more than Rs 1,776 crore in just the first nine months of 2024 alone.

These are not opportunistic, isolated crimes. They are industrialised operations run from organised fraud compounds, frequently across international borders, using spoofed phone numbers, AI-generated voices, and convincing fake government portals to manufacture legitimacy and fear.

Counterfeit currency compounds the challenge. The RBI's Annual Report 2025 recorded a rise in seizures of Fake Indian Currency Notes (FICN), with high-denomination Rs 500 counterfeits now of sufficient print quality to defeat manual detection in routine banking operations.

The Core Gap

Law enforcement does not lack evidence after the fact — it lacks intelligence before mass victimisation occurs, and reliable tools to detect threats at the point of contact rather than at the point of complaint. Addressing this requires the convergence of financial transaction intelligence, communication network analysis, physical security screening, and real-time public safety coordination — a genuinely multi-source, multi-agency intelligence problem.

Why This Is an AI Problem

- Scam calls and video sessions generate patterns — voice, script, cadence — that are learnable and classifiable in real time.
- Counterfeit currency has visual micro-signatures that computer vision can verify faster and more consistently than manual inspection.
- Fraud rings leave a trail across accounts, devices, and numbers that only becomes visible through graph-based network analysis.
- Crime concentrates geographically and temporally in ways that geospatial modelling can anticipate rather than merely record.

2. Our Proposed Solution — SentinelAI

SentinelAI is an AI-powered Digital Public Safety Intelligence Platform built to shift law enforcement, financial institutions, telecom operators, and citizens from reactive case investigation to predictive threat neutralisation. Rather than waiting for a complaint to be filed, SentinelAI continuously fuses signals from communications, financial metadata, geospatial data, counterfeit currency imagery, and citizen reports to surface emerging threats while they are still in progress.

The platform is designed as a single Digital Threat Intelligence Command Centre with distinct, purpose-built AI agents — each solving one dimension of the problem — that share intelligence with one another through

a common fusion and risk-scoring layer. This is what allows a suspicious call, a counterfeit note, and a flagged bank account to be recognised as parts of the same criminal operation rather than three unrelated incidents.

Guiding Principles

- Prevention over investigation — intercept fraud before money moves, not after.
- Fusion over silos — every module feeds a shared intelligence graph.
- Precision over noise — citizen-facing alerts are engineered for a very low false-positive rate.
- Evidence by design — every detection generates an audit trail suitable for legal proceedings.
- Reach at scale — accessible through web, mobile, WhatsApp, IVR, and voice, in 12+ Indian languages.

3. What We Are Going to Build

We will build SentinelAI as a full-stack, cloud-deployable web platform (with mobile and WhatsApp reach) comprising eight interlocking modules, a shared AI intelligence layer, and role-specific dashboards for citizens, police, banks, telecom operators, and government agencies. The table below summarises the eight core modules that form the working prototype.

Module	What It Does
Digital Arrest Scam Detection Engine	Classifies live call/video sessions using voice, script, and metadata signals; issues risk scores and pre-transfer alerts to citizens, telecom providers, and police.
Counterfeit Currency Identification Agent	Computer-vision verification of security thread, microprint, watermark, serial pattern, and UV features across all denominations, on mobile and teller-side devices.
Fraud Network Graph Intelligence	Graph AI linking accounts, devices, UPI IDs, and call records to expose organised fraud rings, leader nodes, and money-mule chains as court-admissible packages.
Geospatial Crime Pattern Intelligence	GIS layer mapping complaints, seizures, and hotspots for patrol prioritisation, resource deployment, and cross-district intelligence sharing.
Citizen Fraud Shield	Conversational AI across WhatsApp, IVR, app, and web that assesses suspicious calls, messages, and payment requests in real time, in 12+ languages.
Intelligence Command Centre	Unified dashboard giving police live scam alerts, fraud trends, hotspot maps, and a national threat overview.
AI Case Builder	Automatic assembly of evidence packages — timelines, transaction chains, graph links, and geolocation — exported as court-ready documentation.
Predictive Threat Intelligence	Forecasts emerging scam campaigns, counterfeit distribution routes, and high-risk districts ahead of escalation.

Deliverables for This Hackathon

- A working, demonstrable prototype covering at least the scam detection, counterfeit currency, and command-centre modules end to end.
- A complete system architecture diagram (Section 6).
- A presentation deck summarising problem, solution, and impact.
- A demo video showing the citizen and law-enforcement journeys.

4. Complete Feature List

The feature set below covers every role-facing surface of the platform — citizen tools, law-enforcement operations, financial and telecom dashboards, and the underlying administrative and safety layers — including additional features introduced beyond the base challenge brief to strengthen adoption, trust, and scalability.

4.1 Home & Public Awareness

- Platform overview and live national threat statistics
- Real-time public alert ticker for active scam campaigns
- Public awareness resources and safety guidance

4.2 Citizen Portal

- Real-time fraud risk assessment for calls, messages, and payment requests
- Counterfeit currency scanner (camera-based, instant verdict)
- Guided fraud reporting with direct NCRB portal linkage
- Complaint status tracking
- One-tap Emergency Panic Button — freeze linked accounts, notify police and family, share live location
- Interactive fraud-awareness quizzes and safety tips
- Senior Citizen Mode and full accessibility support

4.3 Law Enforcement Dashboard

- Live incident feed and active scam session monitor
- Fraud network graph visualisation (accounts, devices, mule chains)
- Geospatial intelligence map with heatmaps and patrol suggestions

- AI Case Builder for automatic, exportable evidence packages
- Cross-jurisdiction case linking and investigation workspace
- Resource deployment and nearest-response-unit routing

4.4 Bank Dashboard

- Counterfeit note reporting and denomination-wise analytics
- Suspicious / high-risk account alerts
- Transaction anomaly and mule-account intelligence
- Branch-level fraud monitoring

4.5 Telecom Dashboard

- Spoofed and high-risk number detection
- Spam / scam call clustering
- SIM and device intelligence
- Live call-pattern monitoring and pre-transfer intercept alerts

4.6 Analytics & Reporting

- Fraud growth trend charts and monthly summary reports
- District-wise ranking and comparison
- Model and detection performance metrics

4.7 Conversational AI Assistant

- Multilingual chatbot and voice assistant (12+ Indian languages)
- Document, screenshot, and fake-notice understanding
- Plain-language scam explanation and legal guidance

4.8 Admin Portal

- Role and permission management across agencies
- Model monitoring, dataset management, and audit logs

- API key and threat-configuration management

4.9 Additional Features Beyond the Brief

These extend the platform's protective surface beyond the illustrative examples in the challenge brief:

- Deepfake and AI-generated voice detection
- Fake government website / portal verification
- QR code scam detection with pre-payment warning
- UPI Safety Checker — merchant reputation and complaint history
- SMS fraud detection (OTP, courier, lottery, bank-impersonation scams)
- WhatsApp forward-and-verify scam checker
- Browser extension warning against phishing and fake banking pages
- Anonymous community intelligence and neighbourhood-watch reporting
- AI Awareness Centre with interactive scam-simulation training and a cyber-safety certification

5. Users & Stakeholders

User Group	Primary Needs Served
Citizens	Verify calls & payments, scan currency, report scams, trigger emergency help
Police	Investigate fraud, monitor intelligence dashboards, visualise criminal networks, manage cases
Banks	Detect counterfeits, monitor transactions, flag suspicious accounts
Telecom Providers	Detect scam calls, assess number reputation, issue live warnings
Government Agencies	Monitor national fraud trends, allocate resources, run public awareness campaigns

6. System Architecture

SentinelAI is organised as a layered, service-oriented architecture. User-facing channels (web, mobile, WhatsApp, IVR, browser extension) pass through a unified API gateway into a set of purpose-built platform modules. Each module calls into a shared AI intelligence layer — NLP/LLM, Speech AI, Computer Vision, Graph AI, Geospatial AI, and Agentic orchestration — whose outputs are combined by a central Intelligence Fusion Engine and a Risk Scoring & Decision Engine. Decisions are routed to the appropriate alert channel, and every material detection is captured by the Evidence & Case Builder into an auditable, court-ready record

before being persisted to the secure data layer and, where relevant, shared with external agencies such as NCRB and MHA.

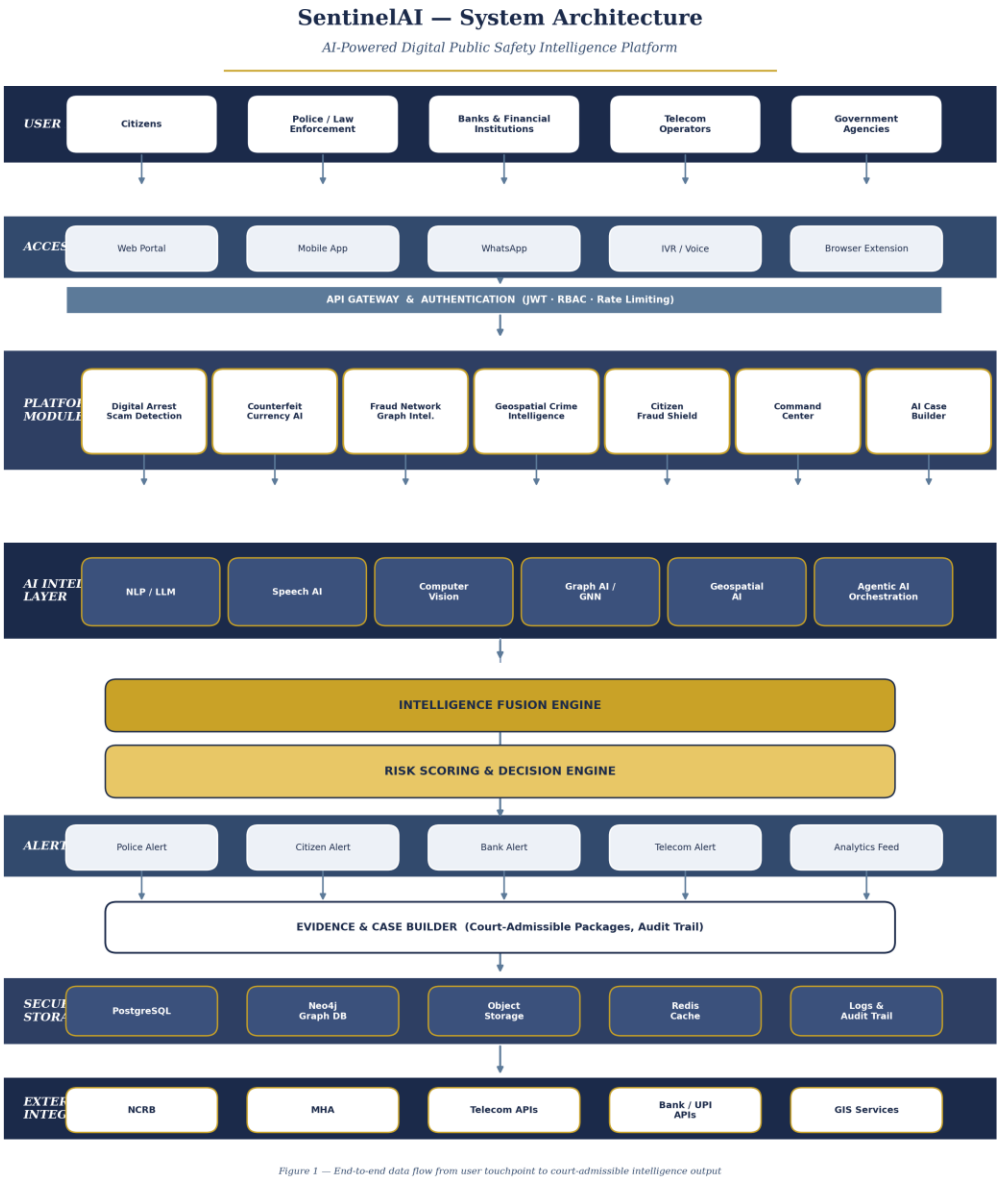


Figure 1 — End-to-end data flow from user touchpoint to court-admissible intelligence output

Figure 1 — End-to-end SentinelAI architecture, from user touchpoint to court-admissible intelligence output.

6.1 Technology Stack

Layer	Suggested Technologies
Frontend	React, Next.js, TypeScript, Tailwind CSS, Mapbox, Leaflet
Backend	FastAPI, Python, Node.js, REST APIs, WebSockets
AI / ML	PyTorch, TensorFlow, OpenCV, Whisper, Transformers, LangChain
Data	PostgreSQL, Neo4j, Redis, Object Storage
Deployment	Docker, Kubernetes, Nginx, GitHub Actions, Cloud Storage

6.2 Security & Compliance

- JWT authentication with role-based access control
- End-to-end encryption and secure API gateway
- Evidence integrity hashing and digital signatures
- Full audit logging and data anonymisation
- Designed for GDPR / DPDP compliance

7. AI Techniques by Module

Module	Core AI Techniques
Digital Arrest Scam Detection	Speech-to-text, NLP scam classification, voice-spoof detection, LLM-based script analysis
Counterfeit Currency Detection	CNN / Vision Transformer, OCR, microprint & UV-feature analysis
Fraud Network Intelligence	Graph Neural Networks, link prediction, community detection (Neo4j)
Geospatial Crime Intelligence	Spatial clustering, heatmaps, predictive hotspot mapping
Citizen Fraud Shield	Multilingual NLP/LLM, intent classification, conversational orchestration
Predictive Threat Intelligence	Time-series forecasting, multi-agent evidence fusion

8. Evaluation Focus & Judging Criteria

Evaluation Focus

- Counterfeit detection accuracy across denominations and print quality
- Digital arrest scam detection precision and recall

- Fraud network detection lead time before mass victimisation
- False-positive rate for citizen-facing tools (must remain very low)
- Auditability of intelligence packages for legal admissibility

Judging Criteria

Judging Criteria	Weight
Innovation	**25%**
Business Impact	**25%**
Technical Excellence	**20%**
Scalability	**15%**
User Experience	**15%**

9. Expected Outcomes & Innovation Highlights

Expected Outcomes

- Fraud intercepted before financial loss occurs
- Reduced counterfeit currency circulation
- Faster police response through unified intelligence
- Organised scam networks detected, not just individual incidents
- Higher conviction rates through stronger, audit-ready evidence
- Improved citizen trust in digital financial services
- Nationwide, cross-agency intelligence sharing

Innovation Highlights

- Multi-agent AI architecture unifying citizens, banks, telecom, and police on one platform
- Real-time digital arrest scam detection ahead of financial transfer
- Computer-vision counterfeit verification across all denominations
- Graph intelligence exposing organised fraud rings and money-mule networks
- Predictive geospatial crime mapping for proactive patrol deployment
- Court-admissible, AI-assembled evidence packages
- Deliberately low false-positive design for every citizen-facing alert
- Multilingual reach across India's linguistic diversity

10. Final Deliverables

- Fully functional web platform with mobile-friendly interface
- AI fraud detection engine (digital arrest scam classifier)
- Counterfeit currency scanner
- Fraud network intelligence dashboard
- Geospatial crime command centre
- AI citizen assistant (multilingual)
- Architecture documentation
- Presentation deck
- Demo video
- Working prototype
- API documentation and deployment guide
- Future roadmap